Department of Artificial Intelligence

భారతీయ సాంకేతిక విజ్ఞాన సంస్థ హైదరాబాద్
भारतीय प्रौद्योगिकी संस्थान हैदराबाद
Indian Institute of Technology Hyderabad
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Reinforcement Learning (AI3000)

Lecture 00: Course Logistics

Karthik P. N.
Assistant Professor, Department of AI
Email: pnkarthik@ai.iith.ac.in

28 July 2026

Timeline

Tue	Fri	Sat	Quiz / TA Session
28 Jul	31 Jul	01 Aug	**T**0
04 Aug	07 Aug	08 Aug	**T**1
11 Aug	14 Aug	15 Aug	–
18 Aug	21 Aug	22 Aug	**Q**1
25 Aug	28 Aug	29 Aug	**T**2
01 Sep	04 Sep	05 Sep	**Q**2
08 Sep	11 Sep	12 Sep	**T**3
15 Sep	18 Sep	19 Sep	**T**4
22 Sep	25 Sep	26 Sep	**Q**3
29 Sep	02 Oct	03 Oct	**T**5
06 Oct	09 Oct	10 Oct	**Q**4
13 Oct	16 Oct	17 Oct	–
20 Oct	23 Oct	24 Oct	–
27 Oct	30 Oct	31 Oct	**T**6
03 Nov	06 Nov	07 Nov	**Q**5
10 Nov	13 Nov	14 Nov	–

T0–**T**6: TA Sessions   **Q**1–**Q**5: Quizzes   No session / holiday

Mid-Sem Exam: 16 **Sep**

Final Exam: 18 **Nov**

Course at a Glance

Course	Reinforcement Learning (AI3000)
Credits	3
Focus	Learning to make good sequential decisions from interaction with an environment
Foundations	Probability, linear algebra, optimization, and foundations of machine learning
Components	Mathematical foundations, algorithms, and implementation exercises

Learning Objective

By the end of the course, you should be able to model sequential decision-making problems and derive, analyse, and implement core reinforcement-learning algorithms

Assessment	Weightage
Quizzes Best 4 of 5 scores will be considered	40%
Mid-Sem Exam	30%
Final Exam	30%

Student Brigade (Teaching Assistants)

Mahvith Akshintala

Deepshikha Urmaliya

Prantik Pramanick

Vedant Prakash Shenoy

U Adithyan

Course Website: `karthikpn.com/#/teaching/Reinforcement-Learning-2026`

Outline

Foundations and Bandits

- Agent–environment interaction
- Rewards, returns, and policies
- Exploration versus exploitation
- Multi-armed bandits

Markov Decision Processes

- Value functions and Bellman equations
- Policy evaluation and improvement
- Value and policy iteration

Model-Free Learning

- Monte Carlo methods
- Temporal-difference learning
- SARSA and Q-learning

Approximation and Deep RL

- Function approximation
- Policy-gradient methods
- Actor–critic methods
- Deep reinforcement learning